GNBF5020

Practice Examination 01

Cumulative · Lectures 1–6 · 30 marks · 90 minutes · 8 questions

GNBF5020 · Introduction to Molecular Biology and Genetics

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MSc in Genomics and Bioinformatics · Lectures 1–6 · 12 Sep – 31 Oct 2026

Format: Part A 5 × MCQ (10 marks) · Part B 2 × pipeline mapping (12 marks) · Part C 1 × exception handling (8 marks) · 90 minutes

Built strictly from the course materials in this folder: the Lectures 1–5 revision pack, the Lecture 6 guide, the case study bank and the exam focus prediction

Part	What it asks for	Questions	Marks
A	Multiple choice — conceptual, single best option	5	10
B	Pipeline mapping — trace a process from input to output	2	12
C	Exception handling — find where the system breaks	1	8
	Total	**8**	**30**

[ENGLISH ONLY] ALL ANSWERS MUST BE IN ENGLISH. Dr. Wong said it three times across the lectures: *"every assignments and exam should be done in English, even this is a Chinese University of Hong Kong,"* and *"you have to answer in English, although this is a CUHK, haha, okay."* He said he gave zero marks to students who answered in Chinese. Keep gene, protein, pathway, method and drug names in English inside your English sentences.

[HOW TO ANSWER] Show the chain, not the conclusion. Every short answer uses four moves: **1.** the observation, with the number if a number was supplied; **2.** the mechanism that produces it — this is where the marks are; **3.** the clinical or experimental meaning; **4.** the action you would take, or the limit you cannot conclude past. A conclusion with no chain earns partial credit; a chain with a conclusion earns full credit.

[NO ANSWER KEY] This paper carries no answer key. The key and a full post-mortem — for every question you get wrong, a translation of the biological error into computational or data-flow terms — are returned once your answers are submitted. Sit the paper closed-book first.

[WHAT THIS PAPER IS] Deliberately beyond the Midterm Drill Pack. The five question types Dr. Wong named out loud — the Kaplan–Meier curve, the contractile ring, NHEJ→SCID, the mitochondrial pedigree and the lac operon — stay in the Drill Pack and the Exam Focus Prediction, where they belong. This paper sits past them: every question is built on mechanism and inference, and the numbers are given to you so that the reasoning, not the recall, is what is being marked.

Instructions

1. Answer in English. Dr. Wong said three times across the lectures that every assignment and exam is answered in English, and that he gave zero marks for answers written in Chinese. Keep gene, protein, pathway, method and drug names in English inside your English sentences.

2. Show the chain, not the conclusion. For every short answer, use four moves:

No.	Move	What it means
1	**Observation**	What the data shows, in one sentence, with the number if a number was supplied
2	**Mechanism**	The pathway or process that produces the observation — *this is where the marks are*
3	**Interpretation**	What it means for the patient, the assay or the experiment
4	**Action or limit**	What you would do next, or what you cannot conclude

3. State what the method cannot do. Every technique has a limit; naming it proves you know the method rather than the slogan.

4. Use the data given. Do not substitute numbers you remember from elsewhere. If a judgement requires an assumption, state the assumption explicitly.

5. Think in systems terms. The course's mechanisms are the systems you already know:

- a metabolic or signalling pathway is a **directed acyclic graph** — find the first node whose output is never consumed downstream, and that is your failure point;
- an enzyme is a **function** with a substrate and a product — ask what it consumes and what it emits;
- a regulatory state is a **logic gate** — ask how many inputs it needs, and whether the gate is open, closed, latched or amplified by feedback;
- a **feedback loop** has a sign — decide whether it stabilises the system or runs away from it.

6. Answer the question that is asked. If the verb is *explain*, naming a mechanism is mandatory; describing the data is only partial credit.

PART A — Multiple choice

(5 questions × 2 marks = 10 marks)

Five conceptual questions. Single best option only — no marks for selecting more than one letter. Each option is there to fail for a specific, identifiable reason: read them as claims, not as labels.

Single best option. Circle one letter per question. No marks for multiple selections.

Q1. A repair defect, a drug class, and a trap

A research group sequences a colorectal tumour and finds a homozygous loss-of-function mutation in the **exonuclease domain of DNA polymerase ε (Pol ε)**, the leading-strand polymerase. The tumour is **microsatellite stable**, and all four mismatch-repair proteins (MLH1, MSH2, MSH6, PMS2) are **retained by IHC**. The group must now choose a systemic therapy.

Which option is most consistent with the mechanisms taught in this course?

A.	A PARP inhibitor, because the defective exonuclease leaves single-strand breaks unrepaired at the replication fork, so PARP becomes trapped on the DNA and the cell cannot survive.

B.	A topoisomerase-inhibitor antibody–drug conjugate, because a polymerase that cannot proofread is also unable to repair the DNA lesions that the conjugate's payload creates.
C.	An anti-PD-1 antibody, because loss of proofreading raises the spontaneous mutation burden, which generates abundant neoantigens and makes the tumour immunogenic ("hot tumour") and responsive to checkpoint blockade.
D.	No targeted therapy should be attempted, because a loss-of-function mutation in a DNA polymerase makes the tumour undruggable, as was historically true for K-Ras.
E.	A PARP inhibitor combined with an anti-PD-1 antibody, because the neoantigen response requires the mismatch-repair-deficient (MSI-high) phenotype, and this tumour is microsatellite stable.

Q2. The replication fork is a pipeline

Treat the replication fork as a pipeline of ordered operations. A cell line is studied and shows the following:

- **DNA synthesis proceeds at a normal rate** (measured by nucleotide incorporation);
- **Sequencing of the newly made DNA shows many single-strand nicks, all of them on the lagging strand, concentrated at the junctions between adjacent fragments**;
- **No RNA nucleotides are retained in the newly made DNA** (the new strands contain no residual primer sequence);
- **ATM/ATR signalling is elevated, but the cells neither arrest permanently nor die.**

Which defect best explains this pattern, and what do you predict for the leading strand?

A.	Primase deficiency. The lagging strand cannot be initiated at all, so each fragment is missing and the fork stalls at every initiation point.
B.	Deficiency in DNA ligase. Okazaki fragments are not sealed into one continuous strand, so nicks accumulate precisely at fragment junctions; the leading strand, synthesised continuously as a single piece, is essentially unaffected.
C.	Deficiency in topoisomerase. Torsional stress builds up ahead of the fork, so the polymerase falls behind and the nascent strands are left nicked behind it.
D.	Loss of the polymerase's 3′→5′ exonuclease activity. Mismatched bases are never removed, so the nascent strand carries the wrong nucleotide at a fraction of positions.
E.	Deficiency in the machinery that removes RNA primers. Each primer is retained and copied into the DNA, so every junction contains an RNA–DNA transition rather than a clean phosphodiester bond.

Q3. Reading a methylation biomarker before you build the assay

You are developing a blood-based early-detection test for colorectal cancer. You have profiled **CDKN2A promoter methylation** in paired tumour and tumour-adjacent tissue. Results:

Region	Adjacent tissue	Tumour
Exon 1	~17 % methylation	~28 % methylation
Exon 2	~16 % methylation	~41 % methylation

ROC performance of the regions as diagnostic markers:

Marker	AUC
Exon 1	0.847
Exon 2	0.924
Exon 1 + Exon 2 combined	0.938

Which statement is the correct basis for designing the assay?

A.	Exon 2 is the better standalone marker (AUC 0.924 vs 0.847) and the combined exon 1 + exon 2 test discriminates best (AUC 0.938); the choice between regions must be made on discriminative performance, not on the raw methylation percentage.
B.	Exon 1 should be used instead, because it lies closer to the transcription start site and is therefore the position at which methylation is most likely to be functionally silencing the gene.

C.	The combined test cannot outperform exon 2 alone, because both exons are reporting the same underlying methylation state of the same promoter and therefore carry no independent information.
D.	Because tumour-adjacent tissue is biochemically different from truly normal tissue, the tumour-versus-adjacent difference cannot be used as evidence of a tumour-associated mark at all.
E.	The single most informative number to report is the difference in mean methylation (exon 2: 41 % versus 16 %), because it is measured directly on the methylation level itself.

Q4. Two ways to gate a therapy: how many inputs does each gate need?

A company wants a T cell that kills solid tumour cells but spares healthy tissue. Two designs are proposed.

- **Design 1** — express the CAR under the control of a **hypoxia-response element (HRE)**, so the CAR protein is made only where oxygen is low.
- **Design 2** — express the CAR only when the **tumour surface antigen AND a second, tumour-context ligand are both present** (a synNotch-style conditional expression circuit).

Which statement is correct?

A.	Design 1 is an AND gate, because the CAR can fire only where the antigen and the low oxygen signal coincide.
B.	Design 2 is a two-input AND gate — the CAR is expressed only when both inputs are present, so healthy tissue expressing the antigen alone cannot trigger it — whereas Design 1 is a single-input gate whose selectivity depends entirely on the assumption that the relevant healthy tissue is not hypoxic; because roughly half of studied solid tumours contain hypoxic tissue, "hypoxic" is not by itself a marker of tumour identity.
C.	Both designs are AND gates, because hypoxia is a feature that occurs only in tumours.
D.	Design 2 is an OR gate, because either the antigen or the contextual ligand alone is sufficient to license CAR expression.
E.	Design 2 is strictly superior in every setting, so Design 1 should never be used.

Q5. Epigenetic cross-talk: which way does the chain run?

The course describes this cross-talk loop explicitly:

> **hypomethylation of the HDAC promoter → increased HDAC expression → decreased histone acetylation → decreased transcription**

Which statement follows from the same model?

A.	An HDAC inhibitor acts downstream of the methylation mark, so it can restore acetylation and re-express a silenced tumour suppressor; but it does not remove the CpG methyl groups themselves, and the active chromatin state requires both loss of DNA methylation and gain of acetylation.
B.	Hypermethylation of the HDAC promoter should raise HDAC expression, which should raise acetylation, which should raise transcription — because methylation at a promoter always increases expression of that promoter.
C.	Hypomethylation of a promoter generally activates the gene, so the HDAC promoter must behave in the same way as a silenced tumour-suppressor promoter when its methylation falls.
D.	HDAC inhibitors cannot restore transcription, because HDACs act only on non-histone substrates such as p53, HSP90, NF-κB and tubulin.
E.	Because methylation controls HDAC and HDAC controls acetylation, this is a positive feedback loop, and a loop of this kind cannot be broken by a drug.

PART B — Pipeline mapping

(2 questions × 6 marks = 12 marks)

Two long-answer questions. Trace the process from input to output, stage by stage, and say at each stage what it consumes, what it emits, and what it cannot establish.

Trace a biological process from input to output, stage by stage. At each stage name the input, the transformation, the output, and — where the course states it — one thing that stage cannot establish.

Q6. From a tumour bank to a screening assay

Your group has the following serum measurements for **Vimentin**, a candidate marker for hepatocellular carcinoma (HCC), obtained from 152 subjects:

Group	n	Serum Vimentin (ng/mL), mean ± SD	Median (IQR)
Healthy controls	64	176.0 ± 61.3	163.0 (128.3–212.5)
HCC, tumour ≤ 2 cm	44	271.5 ± 165.9	202.5 (160.0–328.5)
HCC, tumour > 2 cm	44	231.3 ± 132.3	179.0 (150.0–264.0)

The candidate was originally nominated from a 2-dimensional electrophoresis study of liver tissue. The study design was: IRB-approved tissue → 2-DE-MS → clinical data + protein identity → statistical and informatic analysis → candidate biomarkers, with validation by qPCR, IHC, Western blot and ELISA.

Answer the following.

(a) Write out the full pipeline that turned this tissue into a serum test, in order, naming the technique at each stage, its input, its output, and one thing that stage cannot establish on its own. Begin with the sample-handling decision and end with the statistical decision that would license clinical use. **(2 marks)**

(b) Identify the **first** stage that fails if the material arriving from the hospital is FFPE rather than fresh-frozen tissue, and state the downstream consequence for every later stage of the pipeline. **(1 mark)**

(c) Explain why the **≤ 2 cm** stratum, and not the > 2 cm stratum, is the one that decides whether this assay can work as a *screening* test. Then state what the reported standard deviations imply about the specificity of the assay at any cut-off you choose. **(1 mark)**

(d) State what comparator group you would need in addition to tumour-adjacent tissue before you could claim that the assay detects early disease in the general population, and explain why the adjacent-tissue comparison alone does not license that claim. **(1 mark)**

(e) Classify Vimentin as a **diagnostic**, **prognostic** or **predictive** biomarker, justify your classification in one sentence, and state what additional data would be required to convert it into the other kind of marker. **(1 mark)**

Q7. From a tumour-infiltrating lymphocyte to a validated antigen

You are building a cancer vaccine from a patient's tumour. You have the tumour, the tumour-infiltrating lymphocytes (TIL) and the patient's blood.

Answer the following.

(a) Write out the pipeline, in order, from the tumour to a dose-ready vaccine. At each stage give the input, the transformation performed, and the output. **(2 marks)**

(b) There are **two independent routes** to the antigen that this pipeline must reach. Name both, state what each one contributes, and explain why having two independent routes makes the antigen call more trustworthy. **(1 mark)**

(c) Identify the validation step that the pipeline cannot skip, state what assay is used at that step, and explain why computational prediction alone does not license you to manufacture the vaccine. **(1 mark)**

(d) Explain why the T cells are treated with **anti-CD3 plus anti-CD28** before sequencing, in terms of what each antibody delivers. **(1 mark)**

(e) State the rule that decides whether the final product is a **personalised** vaccine or an **off-the-shelf** vaccine, and say what you must have demonstrated in each case to justify that label. **(1 mark)**

PART C — Exception handling

(1 question × 8 marks = 8 marks)

One scenario. Find the point where the system breaks, follow the consequence downstream, and say how you would prove where it broke.

One scenario. Trace where the system breaks, what happens downstream, and how you would find it.

Q8. The postdoc who deleted the signal peptide

A laboratory is engineering a CAR for an anti-HER2 T-cell product. The construct is standard: an scFv extracellular domain (derived from the antibody heavy- and light-chain variable regions) → hinge → transmembrane domain → co-stimulatory domain → CD3ζ intracellular signalling domain.

A new postdoc reviews the construct and proposes an "optimisation": delete the N-terminal **signal peptide**, reasoning that "the CAR already has a transmembrane domain, so the signal peptide that targets the ER is redundant — and the protein is going to sit in the membrane anyway."

The postdoc's revision is adopted, the construct is re-made, and the following results come back:

Observation	Result
Transduction of primary human T cells	Efficient; cells survive and expand
CAR mRNA by qPCR	Normal level, comparable to the parental construct
CAR on the cell surface, by flow cytometry	**Not detected**
CAR protein inside the cell, by whole-cell lysate Western blot	Detected, but at a lower level than the parental construct
Antigen binding	The scFv domain produced in *E. coli* binds HER2 normally on its own
Functional assay	The cells do **not** kill HER2-positive targets

Answer the following.

(a) Trace the pipeline from translation of the CAR mRNA to tumour-cell killing, stage by stage, and mark **exactly where it breaks**. Name every stage you would expect to require ER entry, and explain why the failure at that stage produces each downstream observation in the table above. **(3 marks)**

(b) In one or two sentences, state why the postdoc's argument is wrong — that is, why a transmembrane domain does not substitute for a signal peptide. **(1 mark)**

(c) Rank the **two** most likely failure points given the data in the table, and name **one experiment** that would discriminate between them. **(2 marks)**

(d) Name one class of therapeutic protein that this same defect would abolish entirely, and explain why. **(1 mark)**

(e) State the general rule this case establishes about any experiment in which a recombinant protein is expressed, in one sentence. **(1 mark)**

When you have finished

Submit your answers. The **answer key and a full post-mortem** — including, for every question you get wrong, a translation of the biological error into computational or data-flow terms — will be returned once your answers are in.

A note on marking. Part A is worth 10 of the 30 marks and is close to pure recognition. The 20 marks in Parts B and C are where the grade is won, and they are won by the chain, not by the conclusion. A conclusion with no chain earns partial credit; a chain with a conclusion earns full credit.

Source basis: `GNBF5020_Complete_Revision_All_Lectures.md` *(Lectures 1–5, audio + slides),* `GNBF5020_Lecture_6_Exam_Focus_and_Learning_Guide.md` *(Lecture 6),* `GNBF5020_Case_Studies.md`*,* `GNBF5020_Exam_Focus_Prediction.md`*. Every fact, number and quotation in this paper is taken from those course materials. Questions are original to this paper and deliberately sit beyond the Midterm Drill Pack.*